22N31A6684

Mobile Computing

Q. Explain the mobile computing paradigm, its promises, & impediments

Mobile computing enables users to access computing resources, data, and services anytime and anywhere using portable computing devices connected through wireless communications networks.

- Mobile computing supports :
  - -> User mobility
  - -> Device mobility
  - -> Service continuity
- Components of mobile computing
  - -> Mobile Devices
  - -> Wireless Networks
  - -> Mobile Software

Characteristics

- Location independence
- Wireless connectivity

Promises of mobile computing

- Users can access information & services regardless of location
- Supports remote work & real time decision making.
- Applications adapt based on user location, behavior & preferences.
- Ensures continuous connectivity across different networks.

Impediments

- Resource constraints - battery, storage
- Wireless Network Limitations - latency, bandwidth.
- Security & Privacy issues
- Mobility management
- Data Synchronization & Consistency

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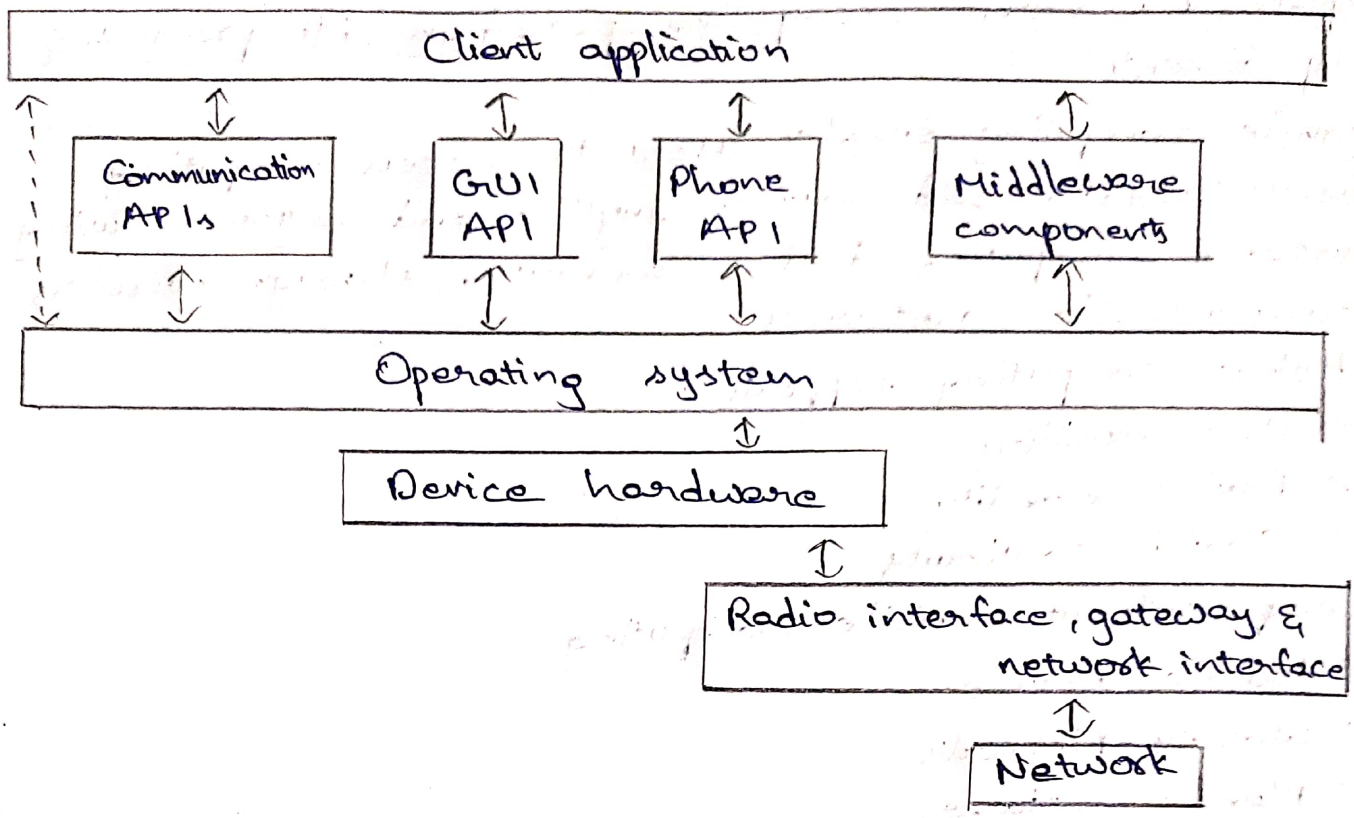
Q. Describe the architecture of mobile communication system.



External Networks → Core networks → Radio Access Network → Mobile station

Mobile Station / User equipment

- Includes mobile device and SIM/USIM
- It's responsible for:
  - → Voice & data communication
  - → Signal transmission & reception
  - → User authentication
- Functions: Radio signal processing
  Mobility detection
  Encryption & decryption.

Radio Access Network (RAN)

- Provides wireless interface between mobile device & network.
- Functions: Radio resource management
  Handover control
  Signal modulation / demodulation

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Core Network

- Handles switching, routing, mobility & service control
- Functions : Session management
  Authentication & authorization

Databases & Support nodes

- These store subscriber & network information.

External networks

- Provides access to emails, web services, video streaming & calls.

Q. Discuss location management & registration in mobile networks.

- Location management is process by which a mobile network keeps track of current location of user to deliver incoming calls & data services efficiently.
- Ensures seamless mobility across areas.
- Consists of 2 main operations : location update & paging.
- Registration - mobile station informs network about its current location when in new area.
- During registration, mobile station sends a location update request to the base station, which is forwarded to core network.
- Network updates user location in databases such as Home Location Register (HLR) & Visitor Location Register (VLR)
- Paging is process by which network searches for mobile device within a location area when there's incoming data session.
- Challenges include increased signaling traffic, higher power usage, & scalability issues in densely populated network.

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Q. Explain SDMA, FDMA, TDMA & CDMA with their advantages.

- SDMA (Space Division Multiple Access)
  Users are separated based on physical location or direction using cell sectoring or directional antennas.
- Advantage - Increases capacity & reduces interference.
- FDMA (Frequency Division Multiple Access)
  The available spectrum is divided into distinct frequency bands, with each user assigned a separate band.
- Advantage - Simple design & low transmission delay.
- TDMA (Time Division Multiple Access)
  Multiple users share the same frequency by transmitting in different time slots within a frame.
- Advantage - Better spectrum utilization & flexible resource allocation.
- CDMA (Code Division Multiple Access)
  All users transmit over same frequency and time, distinguished by unique spreading codes.
- Advantage - High interference resistance & support for soft handoff.
- These multiple access techniques enable efficient sharing of limited radio resources in mobile communication systems.

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Q. Explain problems of conventional TCP in mobile networks.

- Conventional TCP is designed for wired networks & assumes that packet losses occur mainly due to network congestion, which is not always true in mobile networks.
- Packet losses often occur due to wireless errors.
- Responds to packet loss by reducing congestion window.
- High bit error rates
- Mobility & handoffs.
- Variable & asymmetric bandwidth.
- Frequent route changes.
- Long & variable round-trips times
- Power & bandwidth constraints.

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